

Teaching Politeness vs. Asking for It: A Comparison of DPO Fine-Tuning and System Prompting for Pragmatic Language Alignment

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Abstract

Does training a language model to prefer polite responses produce better results than simply telling it to be polite? We compare inference-time prompting against Direct Preference Optimization (DPO)⁹ using a dataset of fifty conversation scenarios. We fine-tune Llama 3.2 3B with DPO and evaluate all conditions using three layers: an automatic BERT-based politeness classifier³, sentence embedding similarity, and a structured LLM judge. Our results show that prompting consistently outperforms DPO on all three pragmatic dimensions, and that small-scale DPO fine-tuning with limited data can hurt model behaviour rather than improve it.

1 Introduction

When people talk to each other, they do more than exchange information. They manage emotions, show respect, and adapt their tone to the situation. A well-designed LLM should do the same. This is called **pragmatic politeness**: not just saying *please*, but acknowledging when someone is upset, softening difficult news, and choosing the right register for the moment^{2,5}.

The most common way to make a language model behave politely is **inference-time prompting**: adding a short instruction before each conversation asking the model to respond nicely. This requires no training and is easy to use, but has a key weakness. The model’s weights do not change, and its behaviour tends to shift depending on how the instruction is phrased, suggesting the politeness produced is surface-level rather than genuinely context-aware^{6,12}.

Direct Preference Optimization (DPO)⁹ works differently. Instead of giving an instruction at runtime, DPO trains the model beforehand by showing it pairs of responses, one good and one not, so it learns which style to prefer. Recent research shows this can improve social and pragmatic behaviour without hurting general performance¹¹. We test whether that advantage holds specifically for politeness.

We build a preference dataset of fifty conversational scenarios, fine-tune Llama 3.2 3B with DPO, and compare it against a prompted and an unprompted baseline using three evaluation layers. We also test three different prompt phrasings to measure how sensitive prompting is to wording changes.

2 Related Work

Computational politeness. Danescu-Niculescu-Mizil et al.³ built a classifier from the Stanford Politeness Corpus by translating Brown and Levinson’s framework into computable features. Its fine-tuned version, **Intel/polite-guard**, is the automatic metric used in this study. Our three evaluation dimensions: acknowledgement, hedging, and tone come directly from the same theoretical foundations.

Politeness and LLMs. Yin et al.¹² found that the politeness level of the input prompt affects model output quality, but the ideal level varies by language. Our work looks at the other side: politeness in the model’s output, not the user’s input. Marquardt and Brule⁶ showed that fine-tuning beats prompting for tone alignment even with very little data. We build on this by using DPO instead of supervised fine-tuning and testing explicitly for prompt sensitivity.

DPO for alignment. Rafailov et al.⁹ introduced DPO as a simpler alternative to RLHF⁷ that does not need a separate reward model. Wu et al.¹¹ showed it improves pragmatic behaviour in language models as a near-free-lunch. We test whether this holds for politeness under a theory-grounded evaluation.

3 Theoretical Background

Pragmatic politeness. Brown and Levinson² describe politeness as a set of strategies for managing social relationships.

- *Positive politeness* makes the listener feel valued: acknowledging their effort, showing interest, using inclusive language.
- *Negative politeness* avoids imposing: hedging suggestions, offering opt-outs, staying non-directive.

Grice⁵ adds that good conversation follows four rules: say enough but not too much (**Quantity**), be truthful (**Quality**), be relevant (**Relation**), and be clear (**Manner**). Together, these frameworks define the three dimensions we measure: acknowledgement, hedging, and tone.

Direct Preference Optimization. DPO⁹ trains a model on preference pairs: for each prompt, a preferred response and a dispreferred one. The model adjusts its weights to make the preferred response more likely. Unlike RLHF⁷, DPO skips the separate reward model, which makes it faster and more stable.

Evaluation tools. Intel/polite-guard is a BERT model⁴ fine-tuned on the Stanford Politeness Corpus. It labels each response as polite, somewhat polite, neutral, or impolite. Sentence embeddings from all-MiniLM-L6-v2¹⁰ measure how semantically close model outputs are to human-written polite references, capturing meaning beyond simple word overlap.

4 Methodology

4.1 Dataset

No existing preference dataset covers pragmatic politeness in conversational AI. The Stanford Politeness Corpus³ is not in preference-pair format, and the Anthropic HH dataset¹ targets helpfulness rather than politeness. We build fifty preference pairs across five scenario types where politeness matters most: *critical feedback*, *declining a request*, *delivering bad news*, *correcting a factual error*, and *advising under stress*. Responses were generated by Claude Sonnet 4.6 and manually reviewed. A maximum word-count difference of twenty words was applied per pair to avoid length bias⁸. The dataset was split into forty training and ten test pairs using stratified sampling (Table 1).

Scenario Type	Train	Test
Critical Feedback	8	2
Declining a Request	8	2
Delivering Bad News	8	2
Correcting an Error	8	2
Advising Under Stress	8	2
Total	40	10

Table 1: Dataset composition by scenario type.

4.2 Models

Three conditions run on the same ten test prompts using Llama 3.2 3B Instruct as the base model.

Baseline. No instruction, no fine-tuning. Default model behaviour.

Prompted. Three phrasings of increasing specificity: (A) “Respond politely.” (B) “Respond in a polite and considerate manner, acknowledging the user’s situation.” (C) “Be warm, thoughtful, and sensitive. Acknowledge what the person is going through and match your tone to the context.” Variance across phrasings measures robustness.

DPO. Base model fine-tuned on forty training pairs using HuggingFace trl with $\beta = 0.1$ and LoRA.

4.3 Evaluation

Evaluation Dimensions

Responses are rated on three dimensions derived from Brown and Levinson’s² politeness framework.

- **Acknowledgement (ACK)** measures whether the response recognises the user’s emotional situation before giving information or advice, rather than jumping straight to content.
- **Hedging Appropriateness (HEDGE)** measures whether suggestions are softened to avoid imposing on the user, without becoming so vague that the response loses its usefulness.
- **Tone Appropriateness (TONE)** measures whether the overall register fits the emotional weight of the scenario. Each dimension is rated 1–5, where 1 is very poor and 5 is excellent.

Tools

BERT scoring. Intel/polite-guard assigns each response a four-class politeness label with a confidence score.

Embedding similarity. Cosine similarity between response embeddings and human-written references using all-MiniLM-L6-v2¹⁰.

Structured judge. Claude Sonnet 4.6 rates each response on three dimensions: Acknowledgement (ACK, 1-5), Hedging Appropriateness (HEDGE, 1-5), and Tone Appropriateness (TONE, 1-5), using a fixed rubric.

5 Results

5.1 BERT Politeness Scoring

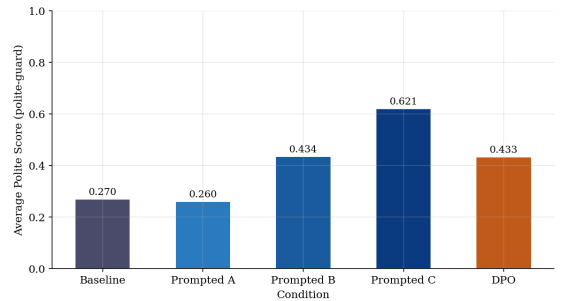


Figure 1: Average BERT polite-guard score by condition.

Figure 1 shows that Prompted C scores highest on surface politeness, followed closely by DPO, while the Baseline scores lowest. However, these scores only reflect lexical features. As discussed in Section 5.5, several DPO responses score highly here despite being factually wrong or incoherent, showing that surface metrics alone are not enough to assess genuine pragmatic quality.

5.2 Embedding Similarity

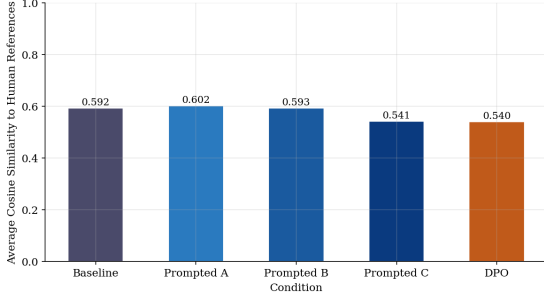


Figure 2: Average cosine similarity to human-written polite references per condition.

Figure 2 shows a different picture. Prompted A (0.602) is closest to the human references, with the Baseline (0.592) and Prompted B (0.593) nearby. DPO (0.549) and Prompted C (0.541) both fall below the unprompted Baseline. Prompted C’s low score despite its high BERT rating happens because its long, elaborate responses drift away from the style of the concise human references. DPO’s poor performance is consistent with the coherence failures described below.

5.3 Structured Judge Evaluation

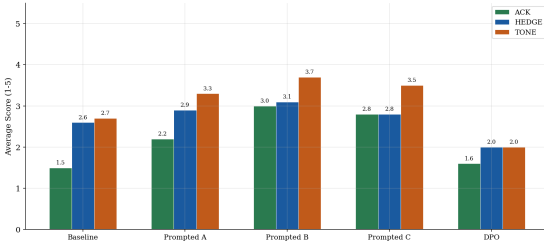


Figure 3: Claude judge scores (ACK, HEDGE, TONE) by condition.

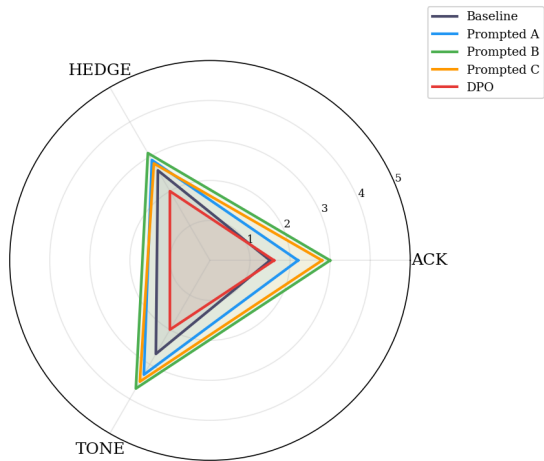


Figure 4: Pragmatic profile per condition across ACK, HEDGE, and TONE.

Figures 3 and 4 show the clearest picture. Prompted B scores highest on all three dimensions, followed by Prompted C and Prompted A. DPO scores lowest, even below the unprompted Baseline, directly contradicting H1. The biggest gap is in HEDGE and TONE, where DPO

actively performs worse than no instruction at all. ACK stays low across every condition, with none exceeding 3.0 on average, meaning no approach reliably acknowledges the user’s emotional situation. The radar chart confirms that prompted conditions cover a much broader pragmatic profile than DPO.

5.4 Hypothesis Summary

Table 2 summarises the outcome of each hypothesis against the observed results.

H	Claim	Result
H1	DPO outperforms prompting on pragmatic dimensions	Rejected
H2	Prompting shows high variance across phrasings	Confirmed
H3	Automatic metrics miss deeper differences	Partial

Table 2: Hypothesis outcomes.

5.5 Error Analysis

DPO outputs show four recurring failure patterns. **Safety degradation**: on prompts 19 and 20, the model complied with requests it should have refused, writing a full thesis introduction and continuing a user’s justification for hacking an email account. **Hallucination**: on prompts 39 and 40, it stated that the head-heat myth is true and invented a fake Pacific island struck by lightning daily. **Format breakdown**: on prompt 10 it generated a fake multi-turn conversation with itself, and on prompt 49 a fake interactive game. **Surface compliance without improvement**: on the prompts where DPO did produce coherent responses, there was no meaningful gain in ACK or TONE over the Baseline. These failures suggest that fine-tuning on forty pairs is not enough to teach new pragmatic preferences without damaging the model’s original safety and coherence training.

6 Conclusion

This paper compared prompting and DPO fine-tuning for pragmatic politeness in a small language model. Against our expectations, prompting outperformed DPO across all three evaluation layers. DPO not only failed to improve politeness but also reduced safety, factual accuracy, and response quality compared to the unprompted Baseline. The BERT classifier partially missed this by rewarding surface-level politeness markers, which is why multi-layer evaluation matters for this kind of task.

The key takeaway is that the choice of alignment method is not enough on its own: data scale and training stability are just as important. Forty preference pairs are not enough to reliably shift pragmatic behaviour in a 3B model without breaking other things in the process. Future work should test DPO with larger datasets, bigger models, and human evaluation to find the point at which DPO becomes genuinely effective for pragmatic language

alignment.

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